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Abstract

Rice production and supply in the Philippines remains to be a significant challenge over the years, and has been facing the concerns brought forth by climate change. In addition, the recent episodes of the El Niño phenomenon have highlighted these concerns on top of the technological, economic, social, and political dimensions of the problem. The recent policy takeaways of the government to impose price caps on rice for distribution in the market has posed serious concerns on the soundness of the policy in terms of its economics. There has also been a move to redistribute excess supply but arguably edible rice, but has been planned to be extended via importation. These policies reflect a myriad of complexities of the agricultural sector that require a comprehensive, long-term and a farm-to-table mode of re-programming for the sector. To complement these assertions, we present and analyze the reported official production, supply, and import data, as well as the metrics of technological requirements, infrastructure, and investment. We also correlate these data given demand, consumption, and nutritional requirements, with rice as a vehicle given its socioeconomic and cultural importance. Given these observations, insights, and inferences, we propose some simple yet significant ways forward to augment the pressing concerns on the sector pos-Covid-19 pandemic onto the "next normal".

Keywords: El Niño phenomenon, rice agriculture, Rice self-sufficiency, Agricultural price caps

Introduction

Rice, being arguably the most important agricultural product in the daily lives of Filipinos, is also one of the most exploited agricultural products, given its multi-sphere nature: agricultural, social, economic, and political. As a country whose food and social functions place rice in the apex of importance, the industry to which it belongs can be viewed as a multistakeholder industry with a myriad of problems and intrinsic complexities.

Rice has been a subject of concern in the last five years—prices, stable supply, importation policy, climate change and its effects on production—to name a few. We look at some data indications based on official sources and infer from these data indications the interactions of various natural, political, social, and economic forces to the trends and future sustainability of the industry, and the rice as an agricultural commodity itself. Our society regards the famous Banaue Rice Terraces—with all of its facts, myths, and problems (e.g., Acabado et al., 2019), and our country is the host to the headquarters of the International Rice Research Institute at the UP Los Baños campus. The government also enshrines the importance of rice via the Department of Agriculture (imminent of its logo), and under it is the Philippine Rice Research Institute and the National Food Authority.

Rice Data and the El Niño Episodes

Data from the Philippine Statistics Authority (PSA) indicated that total production is at an increasing trend in at least the past three decades. Despite the El Niño periods—1997-1998 (van Huysen, 2015), last quarter of 2018 to second quarter of 2022 (PAG-ASA, 2019; PAG-ASA, 2024), and last quarter of 2023 to second quarter of 2024—plus the mobility restrictions brought forth by the Covid-19 pandemic, production has been in the uptrend.

Typically, El Niño results in reduced area harvested and lower yield due to drought conditions and water stress. This can be seen during the 1997-1998 El Niño event where there was a significant drop in both yield (-7.82%) and area harvested (-17.50%), leading to a sharp decrease in production. From 2018 to 2023, certain years likely impacted by El Niño include 2019 and 2022. In 2019, production declined, even as yield increased, as there was a decline in area harvested. Similarly, in 2022, there was a slight decline in all metrics, possibly due to recurring El Niño effects. These fluctuations highlight the vulnerability of palay production to climatic disruptions. However, it is noteworthy that even during these challenging periods, there has been a general increase in the trend for both yield per hectare and area harvested, indicating resilience and adaptation within the agricultural sector.

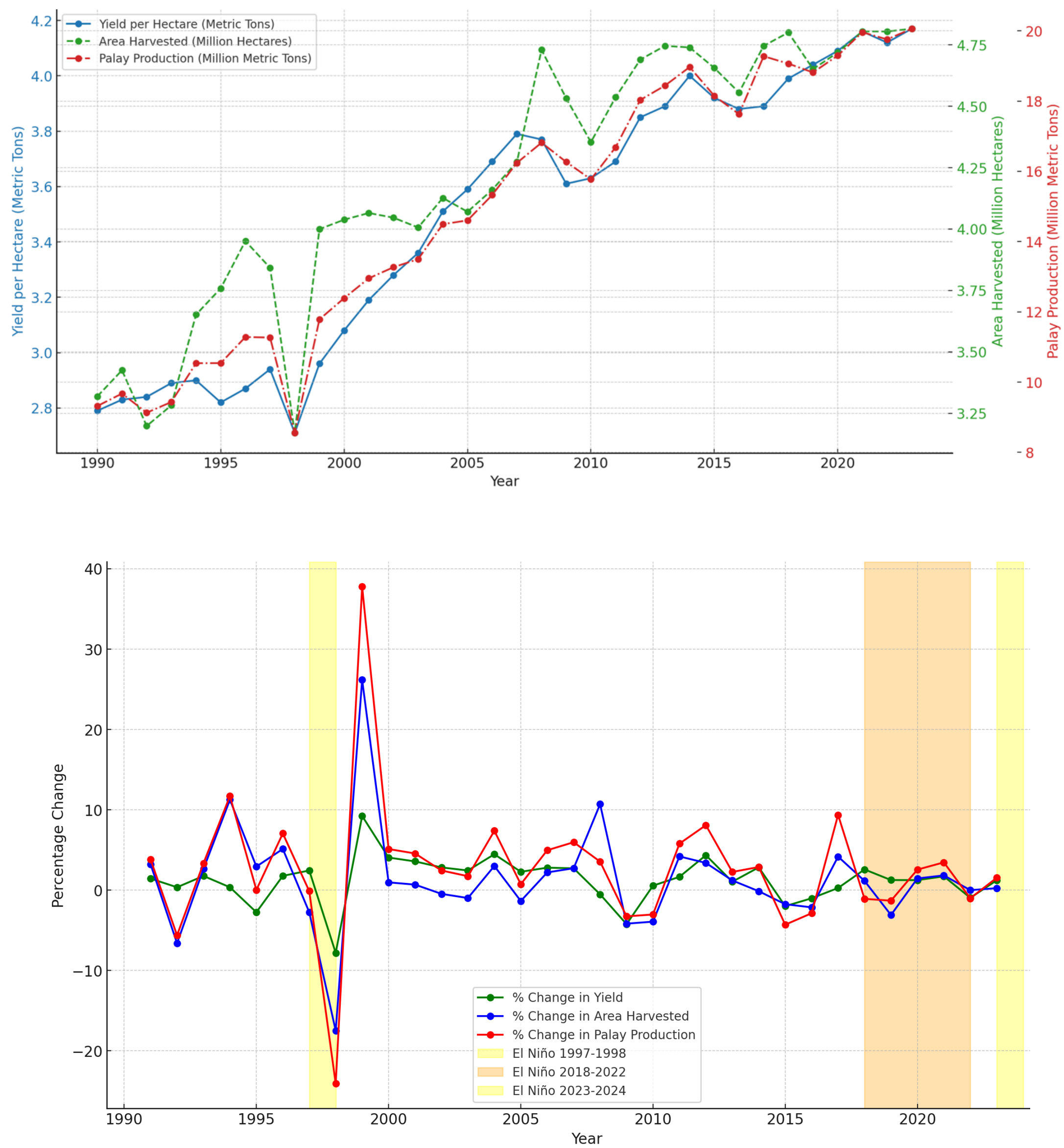


Figure 1. Annual palay production in million metric tons, yield per hectare in metric tons, and area harvested in million metric tons (upper panel), and Percentage change in yield, area harvested and palay production (lower panel), 1990-2023.

Data from the Philippine Statistics Authority.

The relationship between area harvested and yield is clearly illustrated in the figure above (lower panel). Notable increases in palay production occur when both metrics rise simultaneously, while declines in production happen when either or both metrics decrease. El Niño remains a consistent threat to the agricultural sector. Since the available land for cultivation is finite and the area harvested cannot be continuously expanded, it is crucial to focus on improving yield. Enhancing the yield of palay production is essential to counter the impacts of weather disruptions like El Niño.

Figure 2 below is another way to review the production growth in palay. Area growth (2.52%)was the main driver of palay production growth of 2.5% between 1990 and 2000. By 2000-2005, the trend reversed, and production growth came mainly from yield growth of 3.18% (roughly 95% contribution to palay growth). Expansion in area harvested was observed in the period of 2005 to 2015 but only roughly 1.3% growth. However, after this period, area growth has been in decline (0.68% growth between 2020 to 2023). As mentioned earlier, palay production growth will most likely have to come from yield growth. It is therefore important to highlight the vital role of technology and extension support as a pathway for this yield growth. On the other hand, new technologies can only be realized through research and development as well as extension.

It is in this area that the Philippines continues to lag in terms of public investments. According to ASTI, the country invested USD 298 million in 2017 (in PPP and constant 2011 prices). This amount seems substantial but compared to Southeast Asian countries, it is far lower. See for example Thailand (USD 847 million), Malaysia (USD 629 million) and Indonesia (USD 629 million). In terms of research intensity ratio, whose benchmark value should be around 1% of agriculture GVA, the Philippines has a ratio of only 0.41. Clearly there is still room for ramping up investing in R&D. However, there are nuances that must be understood including governance, allocation of research funding, capacities, as well as political support. All these do not necessarily align together and hence the solution of advocating additional funding is not straightforward.

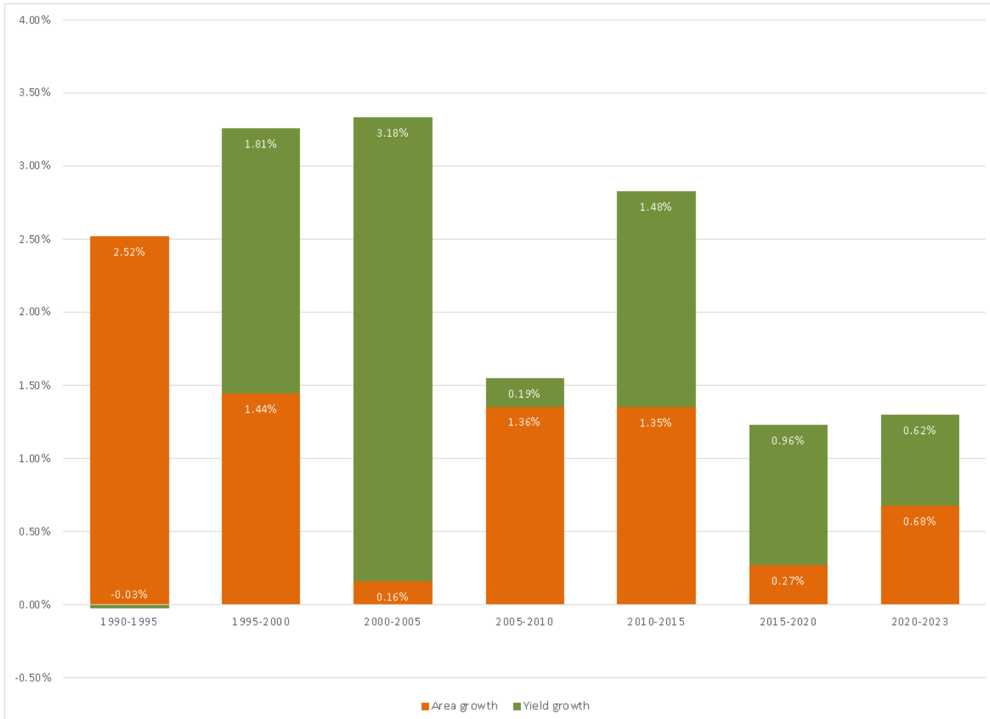


Figure 2. Palay production growth with contributions of area and yield growth, 1990-2023.

Table 1. Agricultural R&D spending by country, 2003, 2013, and 2017[1]

Country	**Million PPP dollars; 2011 constant prices**		
	2003	**2013**	**2017**
Cambodia	16	27.2	30.2
Indonesia	909.6	726.9	629.7
Laos	20.3	16.8	19.3
Malaysia	663.1	657.3	629
Myanmar	6.8	21.6	46.6
Philippines	219.4	195.8	298.4
Thailand	533.6	730.6	847.2
Vietnam	143.6	155.9	177.6
TOTAL	**2,512.3**	**2,532.2**	**2,678.0**

Note. Data lifted directly from the report; From Agricultural Research In Southeast Asia A Cross-Country Analysis Of Resource Allocation, Performance, And Impact On Productivity, Stads et al., 2020.

[1] Constructed by authors from ASTI (various years); 2003 data for Thailand are from Suphannachart (2016); See ASTI's country pages for more detailed information on country-level expenditures for the entire 2000-2017 period (https://www.asti.cgiar.org/countries).

Table 2. Agricultural research intensity ratios by country, 2003, 2013, and 2017[2]

Country	Agricultural R&D spending as a % of AgGDP		
	2000	2013	2017
Cambodia	0.19	0.20	0.22
Indonesia	0.39	0.22	0.17
Laos	0.68	0.28	0.26
Malaysia	1.59	1.04	0.85
Myanmar	0.03	0.03	0.06
Philippines	0.35	0.28	0.41
Thailand	0.75	0.64	0.94
Vietnam	0.17	0.19	0.20
TOTAL	**0.50**	**0.34**	**0.33**

Note. Data lifted directly from the report; From Agricultural Research In Southeast Asia A Cross-Country Analysis Of Resource Allocation, Performance, And Impact On Productivity, Stads et al., 2020.

Source: FROM PAGE 23-24 (STADS, ET AL, 2020)

Rice Farming and Climate Change

As rice production faces the adverse impacts of climate change, it is also undeniable that its production significantly contributes to the emissions that drive climate change. The agricultural sector, particularly rice cultivation, is a notable contributor to greenhouse gas emissions, which exacerbate global warming. Rice production emits significant methane (CH_4), a potent greenhouse gas, which is about 28 times more effective than carbon dioxide in trapping heat (United Nations Environmental Protection Agency, 2023). Flooded rice paddies contribute notably to global methane emissions, comprising 12% of anthropogenic methane emissions and approximately 1.5% of the overall warming effect from greenhouse gases (Asian Development Bank, 2023).

In the Philippines, recent data from PSA shows that rice cultivation emits 782,810 metric tons of methane annually. This significant amount underscores the country's contribution to global methane emissions from rice farming, highlighting the need for sustainable practices to reduce these emissions. Addressing these emissions is crucial for mitigating climate change while ensuring sustainable rice production.

[2] Constructed by authors from ASTI (various years); 2000 data for Thailand are from Suphannachart (2016); The intensity ratios indicate each country's investment in agricultural research as a share of its agricultural gross domestic product.

According to the World Resources Institute (2018), methane emissions from rice cultivation are more closely linked to the area harvested than to the quantity of rice produced. Therefore, increasing rice yield per unit area, rather than expanding the area planted, can lead to a reduction in emissions. Data from 1990 to 2023 shows a significant increase in rice yields per hectare. This continuous trend of increasing yields per hectare can help mitigate methane emissions. However, data also shows that the area harvested has been increasing, with a higher percentage change over the years. This contributes more to the growth of production and might offset the effect of increased yield on climate change mitigation.

Another measure to reduce methane emissions in rice cultivation is employing Alternate Wetting and Drying (AWD). Research from the International Rice Research Institute (2019) indicates that the AWD technique can reduce methane emissions by 30%–70% and decrease water usage by 30%, all without compromising the rice yield for the season.

There are many other ways to address the climate impacts of rice production. Striking a balance between the dual goals of food security and climate change mitigation should always be considered.

To Twenty or Not to Twenty? Or a Bargain to Twenty-nine?

Note that the Covid-19 pandemic has also created a renewed sense of looking at the agricultural sector and its importance (Abueg, 2020; Abueg, 2023). The pandemic, despite its strict mobility restrictions, did not create a dent in the production of rice in the country. However, the recent political spiels beginning the 2022 national elections has subjected the rice industry to the tango of climate change shocks and rice in itself having its own political economy. The intended (versus the experienced) effects of Republic Act no. 11203 (Official Gazette of the Philippines, 2019; Department of Agriculture, 2019) or the Rice Tariffication Law (RTL) did not materialize in the sense that the intent to reduce market prices of rice did not materialize. Possibly, the argued failure of the RTL made possible the win of the current administration through its slogan ticket of promising to reduce market prices of rice to PHP 20 per kilogram. However, towards the middle of the presidential term, President Marcos Jr. argued that it remains to be a goal (Presidential Communications Office, 2023a), and there is false news circulating that belying such promise during the elections (Rappler, 2023).

One short run intervention of the government to mitigate the increasing prices of market rice (which averaged at PHP 60 per kilogram in 2023) was to impose a price cap in market prices that lasted for at least a month (Presidential Communications Office, 2023b; Gita-Carlos, 2023). However, economic theory teaches us that any price control is distortionary, and that to the extreme that it may cause even artificial shortages and irregular markets (e.g., black markets). History also teaches us that these price caps are not new: they are also the same mechanism of the president's father and namesake during the dictatorial years, which led to supply rationing among other problems (Manahan, 2023a; Manahan, 2023b). There are also claims of the government (via the president's vlog) saying that price caps help to stabilize local prices; which are shown to be untrue and are not well-grounded in argument (Vera Files, 2023).

In addition, Speaker Romualdez seconded via his Facebook account that it has influenced world prices to go down (Lalu, 2023), which in many cases is *wrong*. Firstly, the Philippines cannot influence world prices being a small open economy (using the language of macroeconomics). Secondly, related to our economic size relative to other countries and the rest of the world, the Philippines have insignificant share to world supply of rice, and we are even a net importer: the number one importer (Rivera, 2023).[3] And finally, just because there is an indication of correlation does not mean causation, and the previous two arguments may show that this correlation is even spurious (e.g., in Abueg, Macalintal, & Sy Su, 2022).

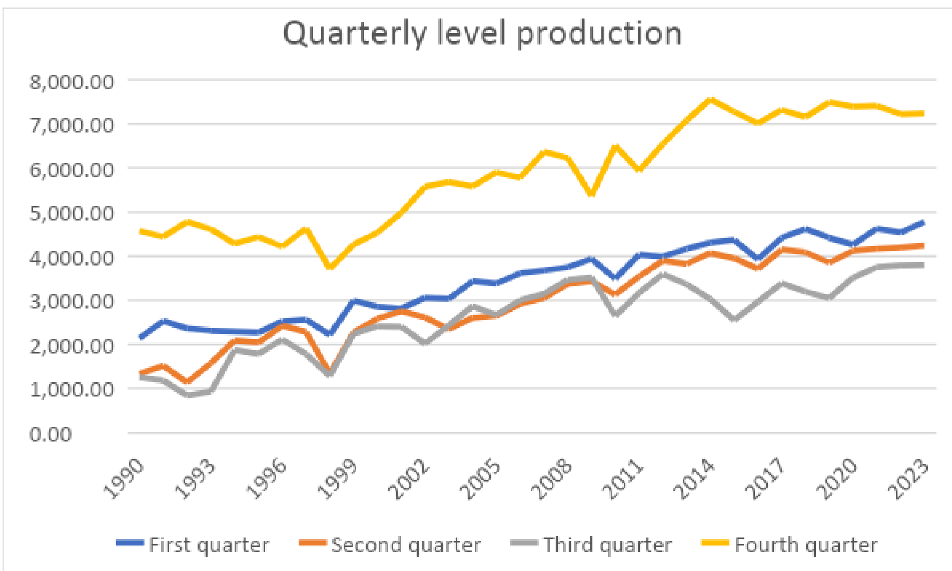
Recent interventions to show commitment by the government on reducing market prices for rice is the mobilization of amendments for the RTL, which is approved by the House of Representatives in the bill's second reading since May 2024. These amendments also aim to address shortcomings of the original RTL since its enactment (Basilio, 2024). Additionally, the Department of Agriculture launched its "Bigas 29" program, which aims to sell "ageing" rice stocks at lower prices. These are rolled out in select outlets particularly in Metro Manila, and targeted to specific groups such as beneficiaries of the 4Ps, solo parents, senior citizens, and persons with disabilities (GMA Integrated News, 2024; Manahan, 2024). While it is in its initial phase, the government puts a limit on individual purchase, but the program will be sustained via importation as announced (Lagare, 2024). These of course, sparks skepticism on the sustainability of the program, and primal on the quality of such "ageing" rice if it is still consumable. Additionally, imports being bought by the government will add to its total bill of expenses, noting that the proposed 2025 national budget will be at PHP 6.352 trillion. Such figure is supported by indications of lowering fiscal deficit by end-2024, and a trajectory of increasing government revenues up until 2028 (Cervantes, 2024; Gita-Carlos, 2024).

[3] These arguments have been echoed by the lead author in the respective previous interviews by ABS-CBN and Vera Files in 2023.

Given these policy interventions, climate change shocks, and other recent economic episodes, we investigate the aggregates of both production and supply utilization data in the next section. Insights from the aggregates will help us provide initial leads and a more holistic perspective on the effects of such interventions and shocks to rice production and to the whole industry.

A Detailed Look into Data Aggregates

Quarterly production data of rice as reported by PSA shows that much of the production of rice in the Philippines is at its peak during the fourth quarter, up to the first quarter of the succeeding year, attributed to the wet season during the second half of the year (Figure 3). These levels dampen as dry season approaches, especially when episodes of drought brought forth by El Nino proceeds (with some local government units declaring a "state of calamity" owing to agricultural destruction of extreme drought. Notably, the recent dry season spelled out high heat indexes, which did affect not only agriculture, but also the supply of potable and irrigation water, and people in the workplace and in onsite classes (Baclig, 2023; Cabato, 2024).



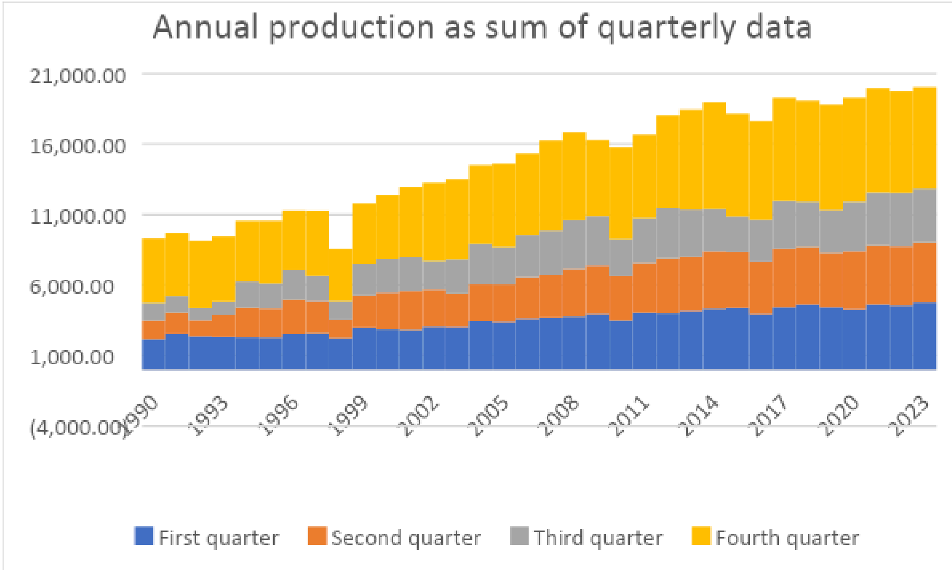


Figure 3. Aggregates of annual production of rice in thousand metric tons (upper panel), and sum to annual data (lower panel), 1990-2023. Data from the Philippine Statistics Authority.

Note also of a "transition" in production in the quarters. Fourth quarter production slowly declines while the first quarter slowly increases, but a relatively steady and similar growth is observed between the second and third quarters. Perhaps, climate change kicks in, since the rainy season is argued to "move" towards the end of the year and continuing to the first quarter of the succeeding year (e.g., see Loo, Billa & Singh, 2015).

For the supply utilization data, local production remains the major source of supply of rice in the country. Current production is supplanted further by a buffer stock that remained unutilized in the previous year. To augment excess demand, the government deems it necessary to import—a facet of the RTL but is aimed at reducing prices and helping revenues generated from tariffs to support local agriculture, especially small-scale farmers. However, in recent years, buffer stocks from a previous year are slowly reducing, thus the augmentation of imports (see Figure 4).

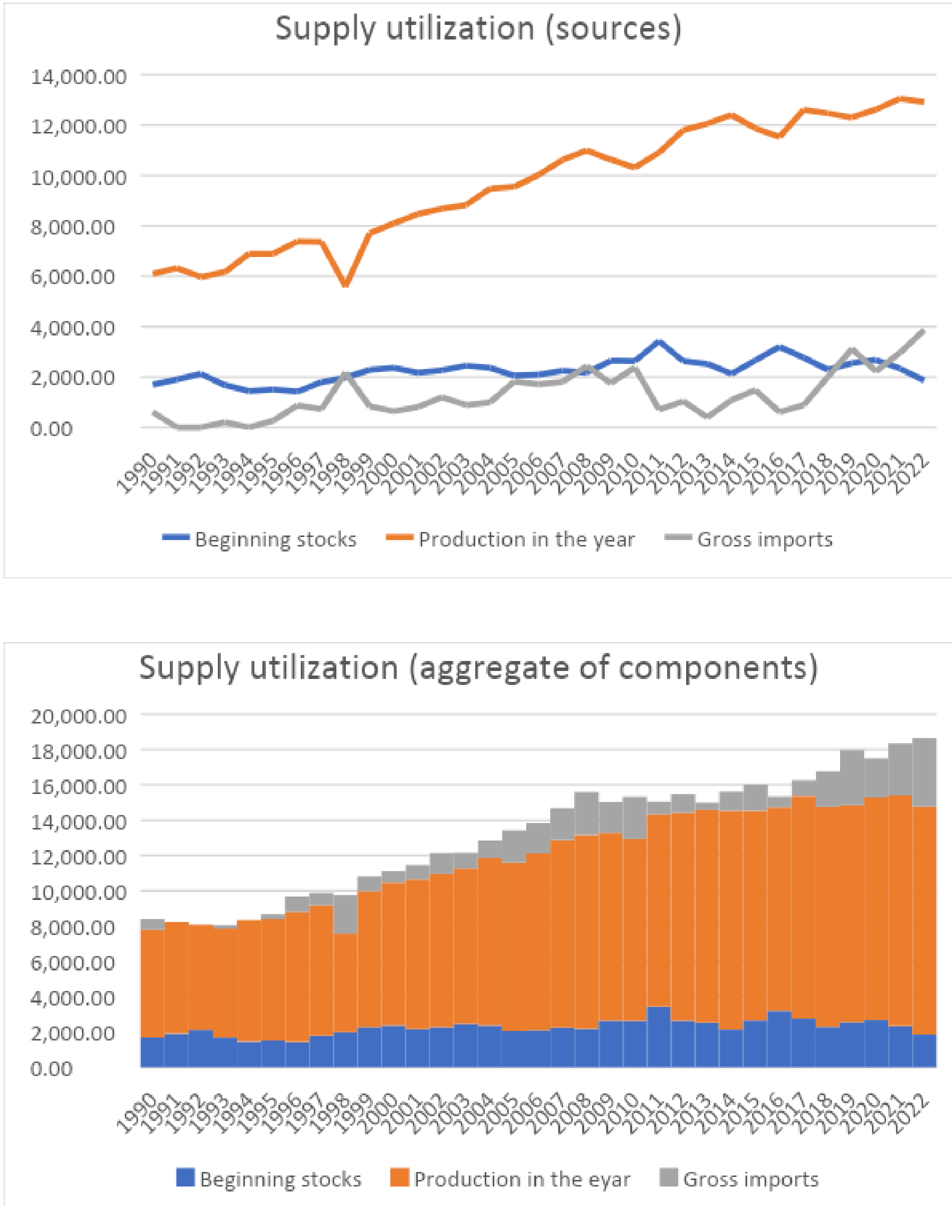


Figure 4. Aggregates of supply utilization of rice in thousand metric tons (upper panel), and sum of aggregates (lower panel), 1990-2022. Data from the Philippine Statistics Authority.

Note also from Figure 4 that local production is slowly plateauing, and with the decline of beginning stocks, importation becomes more and more necessary. However, this has become an issue in addressing the lack of supply of rice in the country. Again, The United States Department of Agriculture noted that the Philippines will remain to be a top importer of rice in the year 2025 (Arcalas, 2024).

With the volatility of demand-supply interactions in the market, Ocampo & Pobre (2021) shows that despite rice inflation being below zero and showing a decline after the RTL passage, the Covid-19 pandemic reversed the trend.

On another note, Crost et al. (2018) have shown correlates of changing environments in agricultural production of rice with local civil conflict. This result provides another angle looking at the centuries-old problem of rice agriculture in the Philippines rooted from colonial times (Corpuz, 1997).

A multitude of studies say that the Philippines has a big room for improving its rice production via modernizing production technology, improving postharvest facilities, among others (Global Rice Science Partnership, 2013).

Conclusions and Ways Forward:

Rice will remain an important part of Filipino diets and thus will remain a dominant crop produced and supported. However, the challenges posed by climate change, dwindling land resources and the limitations in yield growth highlight the need to review and modernize the rice industry.

Declining land areas and a possible reduction in yield growth necessitates that the country continue to invest in R&D to ensure that local rice production can continue to be a major source for domestic demand. The Philippines still lag behind it Southeast Asian neighbors in agricultural R&D spending, underscoring the need for greater public and private sector collaboration to close this gap. While self-sufficiency cannot be sustained, it is important to maintain a degree of national production since recent experiences show that a freer trade regime in what was envisioned by the RTL does not necessarily translate to affordable rice for consumers. Amendments to the RTL and complementary policies should focus on ensuring the benefits of trade are channeled into improving domestic production.

While efforts to increase productivity are very important, there must also be a shift towards sustainable farming practices, particularly in reducing the environmental impact of rice production. Focusing on yield improvements rather than expanding the area harvested is a good approach to reducing the sector's environmental footprint while sustaining production gains.

In the long term, there will be a need to continuously monitor changes in production and demand as the country develops. A question for research and implementing agencies is a transition process towards diversification in production (rice and other crops) and consumption (rice dominated diet to a diversified and perhaps healthy diet). Current thinking is moving to food systems transformation as a means to address a triple challenge of ensuring healthy food, livelihood of producers, and sustainable environment that includes a more hospitable climate in the future.

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